

# TING-WEI HSU

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## RESEARCH INTERESTS

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**RF Sensing, mmWave Systems, Wireless Digital Twin, Datacenter Networks, LLM Serving, Deep Learning for Communications**

## EDUCATION

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**Georgia Institute of Technology**

Atlanta, USA

Ph.D. in Electrical and Computer Engineering

*Aug. 2024 – Present (expected 2029)*

Advisor: [Prof. Karthikeyan Sundaresan](#)

GPA: 4.0/4.0

**National Taiwan University**

Taipei, Taiwan

Master of Science in Communication Engineering

*Sept. 2021 – Feb. 2024*

Advisor: [Prof. Hung-Yun Hsieh](#)

GPA: 4.3/4.3

Thesis Title: Multi-User Pose Estimation Based on Spatial and Temporal Features from WiFi CSI

**Keio University**

Tokyo, Japan

Exchange student

*Sept. 2023 – Feb. 2024*

**National Taiwan University**

Taipei, Taiwan

Bachelor of Science

*Sept. 2016 – Jan. 2021*

Major: Mechanical Engineering

GPA: 3.89/4.3

Minor: Computer Science and Information Engineering

## PUBLICATIONS

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\* Equal contribution.

- [1] **T. -W. Hsu**, K. Sundaresan, “VIP-DT: Semantic Vision Prior for Fast RF Digital Twin Construction,” Under review.
- [2] **T. -W. Hsu\***, Y. -T. Lin\*, K. Sundaresan, “mmTwin: A Scalable Vision-to-RF Digital Twin for mmWave Networks,” Under review. [\[website\]](#)
- [3] Y. -T. Lin\*, **T. -W. Hsu\***, K. Sundaresan, “Predictive Traffic Engineering for Assured mmWave Networking in Industrial AI Applications,” Under review. [\[website\]](#)
- [4] Y. -T. Lin\*, **T. -W. Hsu\***, K. Sundaresan, “Distributed Collaborative Positioning through ISAC for GPS-Denied Robotic Swarms,” Under review. [\[website\]](#)
- [5] H. -C. Lin\*, **T. -W. Hsu\***, C. -E. Ho, A. Saeed, “ARK: Avoiding Routing Collisions for KV Cache Transfer in Disaggregated LLM Inference,” ACM SIGCOMM Workshop on Networks for AI Computing (NAIC), Denver, CO, USA, 2026 (to appear).
- [6] **T. -W. Hsu**, H. -Y. Hsieh, “On Using Spatial and Temporal Features for Robust Multiuser Pose Estimation Based on WiFi CSI,” IEEE Internet of Things Journal, vol. 12, no. 15, pp. 29897–29912, Aug. 2025. [\[paper\]](#)
- [7] **T. -W. Hsu**, H. -Y. Hsieh, “Robust multi-user pose estimation based on spatial and temporal features from WiFi CSI,” 2024 IEEE International Conference on Communications (ICC), Denver, CO, USA, 2024. [\[paper\]](#)
- [8] **T. -W. Hsu** and J. -S. Liu, “Design of smooth path based on the conversion between  $\eta^3$  spline and Bézier curve,” 2020 American Control Conference (ACC), Denver, CO, USA, 2020. [\[paper\]](#)
- [9] **T. -W. Hsu** and J. -S. Liu, “Convex hull determination method for generating Bézier curve followed by Trailer-tractor System,” Automation 2020, Hualien, Taiwan, November 2020. [\[paper\]](#)

## RESEARCH EXPERIENCE

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**Mobile Advanced Research, GaTech***Aug. 2024 – Present*

Graduate Research Assistant

Advisor: [Prof. Karthikeyan Sundaresan](#)

- Built vision-to-RF mmWave digital twins (mmTwin [2], VIP-DT [1]) predicting beamformed link profiles from RGB-D vision, cutting RF site-survey overhead up to  $61\times$  and reaching  $<4$  dB RSSI error on unseen sites.
- Developed AssureNet [3], a digital-twin-driven predictive traffic-engineering framework assuring industrial real-time-AI networking  $\sim 100\%$  of the time (vs.  $\leq 35\%$  baseline), with  $1.3\text{--}1.5\times$  throughput and  $\sim 5\times$  lower disruption.
- Developed DiCoP [4], an infrastructure-free ISAC collaborative-positioning system for GPS-denied swarms, achieving sub-2 m accuracy for  $\sim 89\%$  of nodes and scaling to 100 UAVs.

**Mobile Networks and Wireless Communications Lab, NTU***Sept. 2021 – Aug. 2024*

Graduate Research Assistant

Advisor: [Prof. Hung-Yun Hsieh](#)

- Designed a robust multi-user pose-estimation framework from WiFi CSI, improving cross-domain accuracy by  $\sim 12\%$  (mPCK) over prior methods while running in real time ( $\sim 19$  FPS) [6, 7].
- Proposed three novel approaches to capture spatial and temporal features from WiFi CSI.
- Set up CSI simulation and real experiment platform for experiment and verification.

**Automation Lab, Academia Sinica***Sept. 2019 – Dec. 2020*

Part-Time Research Assistant

Advisor: Prof. Jing-Sin Liu

- Derived  $\eta^3$ -spline / Bézier path transformations and curvature-constrained trajectory design for trailer-tractor parking [8, 9].

**Bio-Inspired Robotic Lab, NTU***Sept. 2018 – Aug. 2020*

Undergraduate Research Assistant

Advisor: Prof. Pei-Chun Lin

- Simulated and analyzed jumping-robot gait patterns in MATLAB.
- Designed and manufactured the robot leg mechanism and tuned its brushless-motor PID controller.

**TEACHING EXPERIENCE****Teaching Assistant**, Computer Communications, GaTech*Fall 2024, Spring 2025***Teaching Assistant**, Computer Programming ( $\times 2$ ) & Probability and Statistics, NTU *2021 – 2022***HONORS AND COMPETITION EXPERIENCE****JASSO Scholarship***Sept. 2023*

- Merit-based scholarship for foreign students, awarded by the Japanese government.

**HITCON World CTF Competition***Jan. 2023*

- Top 40%; responsible for the cryptography section.

**SELECTED PROJECTS****ARK: Collision-Free KV Cache Transfer for LLM Inference**, CS 8803 Datacenter Network Systems, GaTech *Spring 2026*

- Built a packet-level NS-3 simulation of KV-cache transfer for disaggregated LLM inference and designed a distributed source-port path reservation that avoids ECMP hash collisions, cutting P95 flow-completion time up to 34% [5].

**Real-Time Channel Depth Detection Project**, Introduction to IoT, NTUGICE*Fall 2021*

- Built a real-time channel-depth detection system with deep learning, in collaboration with [MiTAC](#).

**SKILLS****Programming:** Python, C/C++, MATLAB,  $\LaTeX$ **ML / DL:** PyTorch, NumPy/SciPy, Vision-Language Models, Neural Radiance Fields (NeRF)**Wireless / RF:** mmWave testbeds, WiFi CSI, USRP / GNU Radio, ns-3, ray tracing**Languages:** Chinese (native), English, Japanese